



Department of Electrical and Electronics Engineering Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3405 Electrical Machines II

Name of the Faculty member (s): Mr. S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic: Unit II / Principle of operation of Synchronous motor**
- **Course Outcome: CO 2**
- **Unit Outcome: 2a**
- **Activity Chosen: Think pair share**
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity: 10 minutes**
- **Details of the Implementation:**

Think-Pair-Share innovative practice conducted for II year EEE students, after explained the concept of principle of operation of synchronous motor. First, I asked the students to think about the reason for not self-starting of synchronous motor for 2 minutes. Then I make them as a pair to discuss their neighbor's and asked the students to discuss for 3 minutes. Finally I asked one of the team to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	2	1	-	-	-	-	-	1	1	-	1	2	-	1

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Principle of operation of Synchronous motor – Think Pair Share
H-Inbaraja [953623105305] K-Thangamalai [953623105057] Reason for synchronous motor is not self starting: * At startup, synchronous motors do not produce sufficient torque to overcome inertia and start rotating on their own. * The stator poles are rotating with synchronous speed and they rotate around very fast and interchange their position. In this case, poles of the stator will attract the poles of rotor and the torque produced will be clockwise. Hence the rotor will undergo to a rapidly reversing torque and the motor will not start.

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

- ❖ **Challenges faced in implementation:**

I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

References:

1. A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, 'Electric Machinery', McGraw Hill publishing Company Ltd, 6th Edition 2017.
2. Stephen J. Chapman, 'Electric Machinery Fundamentals' 4th edition, McGraw Hill Education Pvt. Ltd, 4th Edition 2017.
3. D.P. Kothari and I.J. Nagrath, 'Electric Machines', McGraw Hill Publishing Company Ltd, 5th Edition 2017

Signature of Faculty Member

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RAMCO INSTITUTE OF TECHNOLOGY

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Innovative Practice Description

- **Unit / Topic:** Unit III / Circle diagram- Step by Step procedure
- **Course Outcome:** CO 3
- **Unit Outcome:** 3c
- **Activity Chosen:** Strip Sequence

- **Justification:**

This activity helps students apply what they have learned through reading or teaching. This approach can strengthen students' logical thinking processes and test their mental model of a process. The activity can be done in pairs or groups.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

Strip Sequence activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a Strip Sequence, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	1	-	-	-	-	-	1	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

• Images / Screenshot of the practice:

Innovative Teaching Method Execution	
Circle diagram- Step by Step procedure – Strip Sequence	
Step by step procedure to draw the circle diagram 1) Take the reference power, V as vertical (y-axis) 2) calculate I_{sc} at nominal voltage (I_{SN}) $I_{SN} = \frac{V}{V_{sc}} \times I_{sc}$ 3) calculate W_c at nominal voltage (W_{SN}) $W_{SN} = \left(\frac{I_{SN}}{I_{sc}} \right)^2 \times W_{sc}$ 4) select suitable current scale such that length of I_{SN} between 15 cm to 20 cm $\Rightarrow$ ex: 1 cm = 10 A $I_{SN} = 80 \text{ A}$ 1 cm = 5 A $I_{SN} = 80 \text{ A}$ 1 cm = 80 A $x = 8 \text{ cm}$ $I_{SN} = 80 \text{ A}$ $\checkmark = 16 \text{ cm}$ 5) From No load test I_0, ϕ_0 are obtained draw vector I_0 lagging V by an angle ϕ_0. The line OO' $\phi_0 = \cos^{-1} \left(\frac{W_0}{\sqrt{3} V_0 I_0} \right)$ $\phi_{sc} = \cos^{-1} \left(\frac{W_{sc}}{\sqrt{3} V_{sc} I_{sc}} \right)$ 6) Draw the horizontal line through extremity of I_0 (i.e. O') parallel to horizontal line. 7) Draw the current I_{SN} calculated from I_{sc} with the same scale, lagging V by angle ϕ_{sc} from the origin O. This power is 8 A 8) From 8 A the line 8 A is output line. 9) Draw a I_0 extension of 8 A. extend it to meet line 8 A at point x. This is the centre of circle. 10) Draw the circle, with x as centre and radius equal to $O'x$. This meets the horizontal line drawn from O to 8 A. 11) Draw the I_0 from point A on the horizontal axis, to meet 8 A line at F and meet horizontal axis at D. 12) Torque line The torque line separates stator and rotor cu losses. vertical distance AD represents = Rotor input at short circuit ($W_{SN} V_{SN} = \text{stator cu loss} + \text{rotor cu loss} + \text{rotor cu loss}$) $AF = \text{stator cu loss} + \text{rotor cu loss}$ $\frac{AF}{AF} = \frac{\text{Rotor cu loss}}{\text{stator cu loss}} = \frac{R_2}{R_1} \left(\frac{I_2}{I_1} \right)^2 = \frac{R_2}{R_1}$ Power scale = $W_{SN} / I(AA)$ W/cm Location of E (shy ring) $\frac{AF}{AF} = \frac{W_{SN} - 3 I_{SN}^2 R_1}{3 I_{SN}^2 R_1}$ Location of E (squirrel cage IM) Stator cu losses = $3 I_{SN}^2 R_1$ Rotor cu loss = $W_{SN} - 3 I_{SN}^2 R_1$	

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ ***Benefit of the practice:***

The benefits of Strip Sequence are a great way for students to make notes on all of the information they receive. It made the students the general steps involved in the process and make the students to implement the procedure for the particular topic.

- ❖ ***Challenges faced in implementation:***

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to arrange the strip for the topic of Circle diagram of 3 phase Induction Motor. I planned the activity for 15 minutes only. But in real scenario it takes 25 minutes.

References:

1. A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, 'Electric Machinery', Mc Graw Hill publishing Company Ltd, 6th Edition 2017.
2. Stephen J. Chapman, 'Electric Machinery Fundamentals' 4th edition, McGraw Hill Education Pvt. Ltd, 4th Edition 2017.
3. D.P. Kothari and I.J. Nagrath, 'Electric Machines', McGraw Hill Publishing Company Ltd, 5th Edition 2017

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Course Code & Title: EE3405 Electrical Machines II

Name of the Faculty member: Mr. S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic:** Unit IV / Speed control technique of three phase Induction motor
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** 4c, 4d
- **Activity Chosen:** Mini map
- **Justification:**

Mini mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mini mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches. For this topic, there are different types of speed control techniques can be brought in to single map for better understanding.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Mini mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mini map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

- In my subject Mini map was conducted for the topic Speed control technique of three phase Induction motor. The students are voluntarily drawn the map.

- **CO – PO / PSO mapping:**

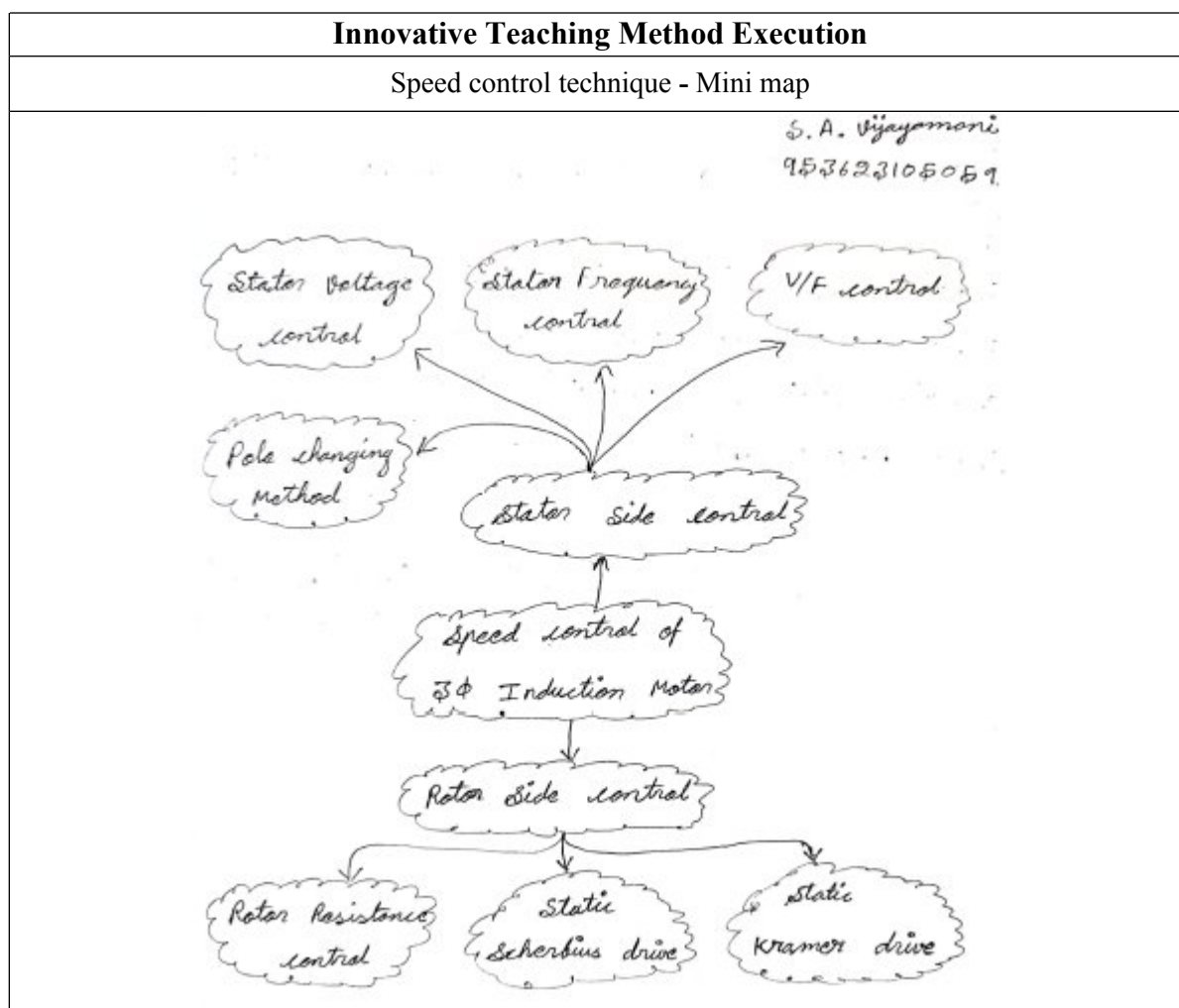
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	1	-	1	-	-	-	1	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The benefits of Mini Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mini map made slow learners to remember the information more quickly.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to indicate relationships between the topics in biomass conversion process. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete

this activity.

References:

1. A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, 'Electric Machinery', McGraw Hill publishing Company Ltd, 6th Edition 2017.
2. Stephen J. Chapman, 'Electric Machinery Fundamentals' 4th edition, McGraw Hill Education Pvt. Ltd, 4th Edition 2017.
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Innovative Practice Description

- **Unit / Topic:** Unit V / Starting methods of single-phase induction motors

- **Course Outcome:** CO 5

- **Unit Outcome:** 5c

- **Activity Chosen:** One-minute paper

- **Justification:**

One-minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

- **Time Allotted for the Activity:** 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	2	1	-	-	-	-	-	1	1	-	1	2	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Starting methods of single-phase induction motors – One minute paper
Starting methods of single phase induction Motor: * Split phase Induction Motor:- * Capacitor split start Induction Motor. * Capacitor run Induction Motor. * Capacitor start-run Induction Motor. * Shaded pole Induction Motor. Arun/Korwar.M 953623105003

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ Students can able to attend the question even in the questions are in indirect form.
- ✓ Students can able to explain the concepts in examination without any confusion.

❖ **Challenges faced in implementation:**

- ✓ Time utilization for conducting activity.

References:

1. A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, 'Electric Machinery', Mc Graw Hill publishing Company Ltd, 6th Edition 2017.
2. Stephen J. Chapman, 'Electric Machinery Fundamentals' 4th edition, McGraw Hill Education Pvt. Ltd, 4th Edition 2017.
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